

Responsible & Trustworthy AI — Learning Resources

Syllabus

The module runs over four sessions, each pairing a concept with runnable code, with a hands-on engineering track running alongside.

- **Session 1 — Foundations of Responsible & Trustworthy AI.** Core principles (fairness, transparency, accountability, robustness, privacy, reliability); trustworthiness vs. accuracy; the **NIST AI RMF** and the **EU AI Act / GDPR**; the AI lifecycle.
- **Session 2 — Explainable AI (XAI).** Interpretable vs. black-box models; LIME, SHAP, partial-dependence plots and counterfactuals; global vs. local explanations; what makes an explanation good.
- **Session 3 — Causal AI for Deeper Trust (optional).** Correlation vs. causation; the ladder of causation; DAGs and identification; causal failure modes.
- **Session 4 — Integration, Auditing & Regulation.** Accountability as a mechanism; the evidence pack; post-market monitoring; incident response; go/no-go and residual-risk acceptance; red-teaming.
- **Engineering track — Prompt & Agentic AI Safety, plus Testing & Best Practices.** Input/output guardrails, prompt-injection defence, PII and sensitive-data handling, least-privilege permissions and human-in-the-loop control; test strategy, monitoring & observability, deployment and the best-practice checklist.

Questions we will try to answer

The module is organised around four questions. Each is anchored in one session but **cannot be answered inside it alone** — a strong answer reaches across the others. By the end, you should be able to move between them without being told that they connect.

- **What licenses the claim that a system is trustworthy?** What evidence — and whose authority — lets us call a system trustworthy at all?
- **What makes an explanation adequate — and to whom?** Why a *plausible* explanation is not the same as a *true* one, and how "adequate" changes with the audience.
- **When may a system act on a number?** What separates a correlation a model has learned from a cause it can safely act on.
- **Who is accountable once the system is live, and by what mechanism?** How oversight, audit and escalation turn accountability from a promise into a mechanism.

Learning Objectives

By the end of this module, all learners will be able to:

1. **Explain the core principles** of responsible and trustworthy AI — fairness, transparency, accountability, robustness/safety, privacy and reliability — and locate them in real deployments governed by the **NIST AI RMF** and the **EU AI Act**. (Bloom's: Remember/Understand)
2. **Apply explainability techniques** (feature importance, SHAP/LIME, partial-dependence plots, counterfactuals) to interpret a pre-trained model's predictions and state the strengths and limits of each. (Bloom's: Apply/Analyze)
3. **Distinguish correlation from causation** using the ladder of causation (association -> intervention -> counterfactual), read a Directed Acyclic Graph (DAG), and identify where causal reasoning makes a system more trustworthy. (Bloom's: Understand/Analyze)

4. **Implement input and output guardrails** — PII detection/redaction, prompt-injection detection, sensitive-topic classification, and schema plus groundedness validation — and explain why each layer exists. (Bloom's: Apply)
5. **Enforce least-privilege tool permissions and human-in-the-loop control** in an agentic system, and estimate the blast radius of a runaway agent. (Bloom's: Apply/Evaluate)
6. **Instrument a trustworthy pipeline** with tamper-evident audit logging, monitoring, incident review and red-teaming, and prove its behaviour with automated tests. (Bloom's: Apply/Evaluate)

By the end of the module, some learners will be able to (depth / stretch):

7. **Compare multiple explanation methods** (e.g., SHAP vs. counterfactuals) and evaluate them with fidelity/stability metrics or a small human-subject study. (Bloom's: Analyze/Evaluate/Create)
8. **Build and query a simple causal model** (DAG plus estimand, DoWhy-style) for intervention or counterfactual analysis on a dataset. (Bloom's: Apply/Create)
9. **Critique a live system** against the NIST AI RMF and the EU AI Act, assemble an evidence pack, and defend a go/no-go deployment decision with named owners, tolerances and review dates. (Bloom's: Evaluate/Create)

Contents to cover

Topic 1 — Foundations of Responsible & Trustworthy AI

- **Core principles of trustworthy AI**
 - Fairness (bias detection and mitigation)
 - Transparency
 - Accountability
 - Robustness / safety
 - Privacy (e.g., differential privacy)
 - Reliability
- **Trustworthiness vs. accuracy** — why a highly accurate model can still be untrustworthy
- **Risk management with the NIST AI RMF** — GOVERN -> MAP -> MEASURE -> MANAGE
- **Regulatory frame** — EU AI Act risk tiers and Annex III high-risk categories; GDPR Arts. 15, 22, 33 and 35
- **The AI lifecycle** — development, deployment and end-user serving

Topic 2 — Explainable AI (XAI)

- **Interpretable vs. black-box models**
- **Post-hoc explanation methods** — LIME, SHAP, partial-dependence plots, counterfactuals
- **Intrinsic interpretability** — decision trees, rule lists, monotonic and generalised additive models
- **Global vs. local explanations** — whole-model vs. single-decision
- **What makes a good explanation** — fidelity, stability, human understandability
- **Audience-specific explanations** — the affected person, the operator and the regulator
- **Explanation misuse** — attributions explain the model, not the world; a plausible explanation can increase over-trust

Topic 3 — Causal AI for Deeper Trust (*optional*)

- **Correlation vs. causation**
- **The ladder of causation** — association -> intervention -> counterfactual
- **Structural Causal Models (SCMs) and DAGs**
- **Reading a graph** — confounding, colliders and mediation
- **Identification** — the estimand vs. the estimate; ITT vs. TOT
- **Causal failure modes** — confounding as the default explanation for observational association
- **Why it matters** — causal explanations are more reliable and actionable

Topic 4 — Engineering Track: Prompt & Agentic AI Safety

- Preventing **prompt injection**
- **Vulnerability identification** in LLM applications
- **Data privacy** and governance — what to provide and what not to
- **PII** — what it is and how to handle it
- Handling **sensitive information**
- **Input/output validation** for AI and LLMs
- **Bias detection**
- **Alignment** (see Anthropic's alignment work)
- Handling **safe-request refusal**
- **Hallucinations** and how to detect them
- **Alignment and autonomy control** — human in / on / out of the loop
- **Governance**
- **Data poisoning**
- **Incident management** — who bears the cost when losses occur

Topic 5 — Integration, Auditing & Regulation

- **Accountability as a mechanism** — a named human-oversight role with authority to override
- **The evidence pack** — model card, DPIA, technical documentation, fundamental-rights assessment, post-market monitoring and incident reports, right to explanation
- **Post-market monitoring** — every signal has a threshold, an owner and an action
- **Incident response and the incident loop** — severity, a clock, and feeding fixes back into the test suite
- **Go / no-go and residual-risk acceptance** — escalating risk tolerance to a human for signature
- **Decommissioning triggers** — decided before a crisis, not during one
- **Red-teaming** — adversarial testing of the system

Topic 6 — Testing & Best Practices

- **Test strategy for AI systems** — unit, property-based, integration and load testing
- **Monitoring & observability** — what to log, measure and alert on
- **Deployment practices** for trustworthy systems
- **Architecture decision records (ADRs)**
- **The best-practice checklist** — prompt engineering, data privacy & RAG, agentic safety, testing, monitoring, deployment, ADRs

Resources + Additional Materials

References & further reading

- **NIST AI Risk Management Framework (AI RMF 1.0)** — GOVERN / MAP / MEASURE / MANAGE.
- **EU AI Act** — Annex III high-risk categories, Chapter III obligations, Arts. 27, 72–73 and 86.
- **GDPR** — Arts. 15, 22, 33 and 35.
- Christoph Molnar — *Interpretable Machine Learning*.
- *Causal Inference for the Brave and True* (Matheus Facure) · DoWhy / EconML tutorials.
- **OWASP Top 10 for LLM Applications** — prompt injection, excessive agency, sensitive-information disclosure.
- Anthropic alignment material — RLHF, RLAIIF, Constitutional AI, Responsible Scaling.
- *Trustworthy AI* (module reference book).

Additional materials

- Module plan — *Responsible and Trustworthy AI*.
- Best-practice checklist — the seven-area "never / always" checklist.
- Question banks — core questions (C1–C4), session questions, and the agentic and scenario question sets.